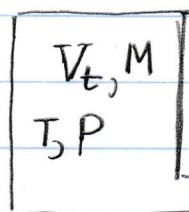


Problem 1. Tank Pressurization Calculations with RK EOS

Consider the following pictorial for the Pressurization Process



V_t = tank volume = 25m^3

T, P = temperature & pressure @ any time

T_f, P_f = temperature & pressure of feed line

m = mass flow rate in, kg/s

M = mass of Nitrogen in tank @ any time

(A) Mass Balance

$$\frac{dM}{dt} = m \quad (1)$$

Energy Balance

Ignoring Potential & Kinetic Energy Contributions we obtain

$$\frac{d(MU)}{dt} = mH_f \quad (2)$$

where U [=] Specific Mass Internal Energy, J/kg

H_f [=] Specific Mass Enthalpy of Feed line, J/kg

Combining (1) & (2) we obtain

$$\frac{d(MU)}{dt} = \frac{dM}{dt} H_f \Rightarrow \boxed{d(MU) = H_f dM} \quad (3)$$

(B) Integrating Eq.(3) we obtain $MU = M_0 U_0 + (M - M_0) H_f$ - (4)

Recognizing that $H = U + PV \Rightarrow U = H - PV = H - P/\rho$

and therefore equation (4) becomes

$$M(H - H_f - P/\rho) = M_0(H_0 - P_0/\rho_0 - H_f) \quad (5)$$

where M_0 = initial mass of N_2 in tank
 ρ_0 = initial density of N_2 in tank
 P_0 = initial pressure of N_2 in tank
 T_0 = initial temperature of N_2 in tank

Also we have $M = \rho V_t$ and $M_0 = \rho_0 V_t$
 and therefore equation (5) becomes

$$\rho \left(H - H_f - \frac{P}{\rho} \right) = \rho_0 \left(H_0 - \frac{P_0}{\rho_0} - H_f \right) \quad (6)$$

Equation (6) can be solved by trial & Error (or using Goal Seek in Excel). The process is described in detail in the steps below:

1. Guess a temperature $T^0 = (\text{say } 50^\circ\text{C})$
2. Using CHE301 Calculator Calculate
 - a. Compressibility factor
 - b. Enthalpy, H
 - c. Density, ρ
3. Substitute values in Equation (6)
4. Repeat [Goal Seek] till equation (6) is satisfied

The steps apply both for IDEAL GAS and for gas following any equation of state. Final Results are shown in the Excel image depicted below:

NUMERICAL SOLUTION**NOTEBOOK**

	IDEAL	RK EOS
Feed Line Temperature, C	25	25
Feed Line Pressure, bar	50	50
Feed Line Compressibility Z	1.0000	0.9874
Feed Line Enthalpy, kJ/kmol	0.0000	-300.3600
Initial Temperature, C	25	25
Initial Pressure, bar	1	1
Initial Compressibility Z	1	0.9996
Initial Density, kmol/m ³	0.040339	0.040355
Initial Enthalpy, kJ/kmol	0.0000	-6.4120
Final Temperature	137.14	127.82
Final Pressure	25	25
Final Compressibility Z	1	1.0033
Final Density	0.732838	0.747393
Final Enthalpy	3274.9416	2926.6726
Equation (See Below)	0.00000	0.00000
Tank Volume, m ³	25	25
Initial Mass, kmol	1.0085	1.0089
Final Mass, kmol	18.3209	18.6848
Mass Added, kmol	17.3125	17.6760

$$\rho \left(H - H_{\text{Feed}} - \frac{P}{\rho} \right) - \rho_o \left(H_o - H_{\text{Feed}} - \frac{P_o}{\rho_o} \right) = 0$$